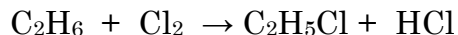
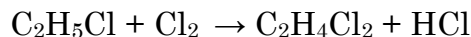


- (1) Ethane is chlorinated in a continuous reactor:



Some of the product monochloroethane is further chlorinated in an undesired side reaction:



- (a) Suppose your objective is to maximize selectivity of the monochloroethane production relative to the dichloroethane production. Would you design the reactor for a high or low conversion of ethane? Explain your answer. What additional processing steps would almost certainly be necessary to make the process economical?
- (b) Draw and label a flowchart, assuming that the reactor feed contains only the two reactants. Use a degree-of-freedom analysis based on atomic species balances to determine how many process variable values must be specified for you to be able to calculate the remainder.
- (c) The reactor is designed to yield a 15% conversion of ethane and a selectivity of 14 moles of $\text{C}_2\text{H}_5\text{Cl}$ for every mole of $\text{C}_2\text{H}_4\text{Cl}_2$. Assume all of the chlorine gas is reacted. Calculate the feed ratio (moles Cl_2 /moles C_2H_6) and the fractional yield of monochloroethane.
- (d) Suppose the reactor is built and started up and the conversion is only 14%. Chromatographic analysis shows that there is no Cl_2 in the product, but another species of molecular weight higher than that of dichloroethane is present. Offer a likely explanation of the results.

(2) Murphy 4.40

(3) Murphy 4.49

(4) Murphy 4.50